

SEMEY, Kazakhstan

WMO: 361770

Lat: 50.417N

Lon: 80.300E

Elev: 196

StdP: 98.99

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.0	-31.4	-37.6	0.1	-35.0	-34.2	0.2	-31.3	9.3	-6.2	8.3	-6.8	1.2	90	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	33.1	19.0	31.4	18.5	29.7	17.8	20.6	29.0	19.8	28.0	19.1	27.1	2.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.3	23.3	17.1	12.5	22.4	16.2	11.8	22.1	60.3	28.9	57.7	28.2	55.2	27.1	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.9	6.7	6.0	-36.0	36.7	4.5	2.0	-39.2	38.1	-41.8	39.3	-44.3	40.4	-47.5	41.9
			-36.0	22.4	4.4	1.2	-39.2	23.2	-41.7	23.9	-44.2	24.6	-47.3	25.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-15.8	-13.6	-4.7	7.4	14.4	20.3	21.6	19.3	12.9	5.7	-4.0	-11.6	(6)
(7)		DBStd	14.48	8.37	8.01	7.28	5.49	4.59	3.69	3.16	3.61	4.57	4.98	7.48	7.46	(7)
(8)		HDD10.0	3301	799	660	456	110	14	0	0	0	23	149	419	671	(8)
(9)		HDD18.3	5354	1057	893	713	328	136	24	9	33	172	392	669	929	(9)
(10)		CDD10.0	1265	0	0	0	33	150	309	359	288	109	16	1	0	(10)
(11)		CDD18.3	278	0	0	0	1	14	83	109	63	8	0	0	0	(11)
(12)		CDH23.3	3479	0	0	0	22	283	977	1157	841	191	9	0	0	(12)
(13)		CDH26.7	1268	0	0	0	4	81	361	448	314	59	1	0	0	(13)
(14)	Wind	WSAvg	2.5	2.4	2.5	2.6	2.7	2.8	2.5	2.3	2.2	2.1	2.3	2.6	2.8	(14)
(15)	Precipitation	PrecAvg	293	14	16	19	18	29	31	49	26	18	26	26	22	(15)
(16)		PrecMax	771	31	42	69	50	118	81	141	74	116	78	83	50	(16)
(17)		PrecMin	145	1	2	2	0	0	2	0	3	0	4	0	3	(17)
(18)		PrecStd	101	7	9	13	13	22	19	34	18	17	16	16	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	2.4	15.2	26.1	32.3	34.4	35.8	35.4	31.7	24.5	14.2	3.2	(19)
(20)			MCWB	0.5	0.4	8.0	14.0	17.1	18.8	20.2	19.6	17.5	12.2	7.1	1.1	(20)
(21)		2%	DB	-0.4	0.7	10.0	22.6	28.7	32.2	32.9	32.4	28.1	20.2	10.2	1.5	(21)
(22)			MCWB	-2.0	-1.2	5.5	12.1	15.8	18.3	19.7	18.8	15.7	10.9	5.7	-0.2	(22)
(23)		5%	DB	-2.4	-0.7	6.6	20.1	26.6	30.5	31.0	30.1	25.3	17.2	7.6	-0.1	(23)
(24)			MCWB	-4.0	-2.3	3.5	11.1	14.8	17.9	19.0	18.1	14.5	9.6	4.1	-1.8	(24)
(25)		10%	DB	-4.4	-2.4	4.0	17.5	24.2	28.7	29.2	28.0	22.5	13.9	4.9	-1.8	(25)
(26)			MCWB	-5.7	-3.9	1.8	9.6	14.0	17.4	18.4	17.2	13.2	8.0	2.3	-3.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	0.9	8.8	14.9	18.3	21.0	22.4	21.1	18.0	12.9	8.6	1.5	(27)
(28)			MCD B	1.8	2.1	14.3	23.6	29.0	30.0	31.6	30.8	29.8	22.0	11.7	2.8	(28)
(29)		2%	WB	-1.9	-0.9	5.8	13.1	16.9	19.9	20.9	19.8	16.2	11.3	5.7	-0.1	(29)
(30)			MCD B	-0.5	0.5	9.4	21.0	26.3	28.1	29.0	29.1	26.1	18.8	9.4	1.4	(30)
(31)		5%	WB	-3.9	-2.3	3.6	11.6	15.7	19.0	20.2	18.9	14.9	9.9	4.1	-1.7	(31)
(32)			MCD B	-2.5	-0.8	6.3	18.9	24.2	27.5	28.0	27.9	23.7	15.6	7.3	-0.2	(32)
(33)		10%	WB	-5.7	-3.9	2.0	10.1	14.6	18.3	19.5	18.0	13.6	8.4	2.4	-3.2	(33)
(34)			MCD B	-4.4	-2.3	3.9	16.7	22.8	26.5	27.0	26.2	21.0	13.2	4.8	-1.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.1	10.5	9.7	12.6	13.8	12.9	12.3	14.0	14.5	11.3	8.2	8.4	(35)
(36)			MCDBR	8.1	8.7	11.3	17.7	17.9	16.0	15.8	18.1	19.4	17.8	11.2	8.4	(36)
(37)			MCWBR	7.5	7.8	7.5	8.9	7.4	5.4	5.2	6.9	8.4	9.4	7.3	7.5	(37)
(38)		5% WB	MCDBR	8.2	8.6	10.8	16.3	15.4	13.8	13.2	15.9	17.5	15.5	10.3	8.5	(38)
(39)			MCWBR	7.7	7.8	7.5	8.6	7.1	5.3	5.0	6.6	8.1	8.7	7.1	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.304	0.344	0.399	0.424	0.424	0.422	0.444	0.414	0.370	0.361	0.346	0.313	(40)	
(41)		taud	2.037	1.955	1.950	2.099	2.148	2.240	2.187	2.246	2.314	2.235	2.112	2.008	(41)	
(42)		Ebn at Noon	681	749	778	814	833	837	810	817	817	739	626	594	(42)	
(43)		Edh at Noon	86	123	149	145	144	133	138	124	103	93	81	77	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.46	2.56	3.88	4.83	6.22	6.61	6.04	5.44	4.07	2.27	1.21	1.03	(44)	
(45)		RadStd	0.10	0.17	0.34	0.37	0.42	0.45	0.41	0.26	0.32	0.27	0.12	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page